

# SQL - Join

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## **Spájanie dvoch tabuliek:**

INNER JOIN

LEFT JOIN

RIGHT JOIN

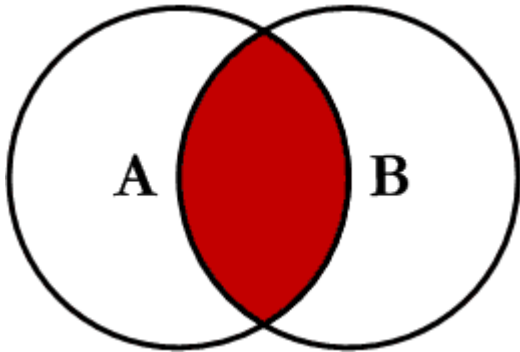
OUTER JOIN

LEFT JOIN EXCLUDING INNER JOIN

RIGHT JOIN EXCLUDING INNER JOIN

OUTER JOIN EXCLUDING INNER JOIN

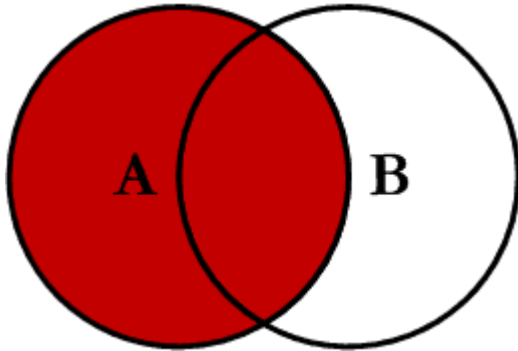
# Inner JOIN



This is the simplest, most understood Join and is the most common. This query will return all of the records in the left table (table A) that have a matching record in the right table (table B). This Join is written as follows:

```
SELECT <select_list>  
FROM Table_A A  
INNER JOIN Table_B B  
ON A.Key = B.Key
```

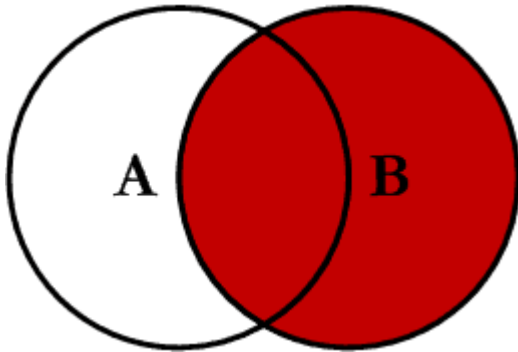
# Left JOIN



This query will return all of the records in the left table (table A) regardless if any of those records have a match in the right table (table B). It will also return any matching records from the right table. This Join is written as follows:

```
SELECT <select_list>  
FROM Table_A A  
LEFT JOIN Table_B B  
ON A.Key = B.Key
```

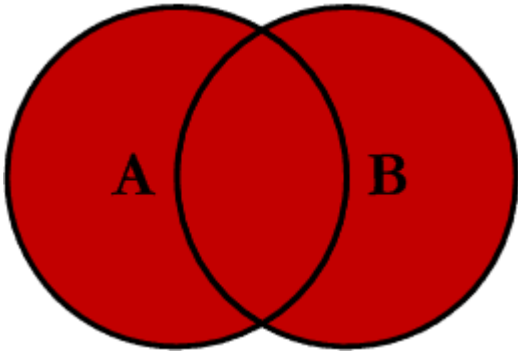
# Right JOIN



This query will return all of the records in the right table (table B) regardless if any of those records have a match in the left table (table A). It will also return any matching records from the left table. This Join is written as follows:

```
SELECT <select_list>  
FROM Table_A A  
RIGHT JOIN Table_B B  
ON A.Key = B.Key
```

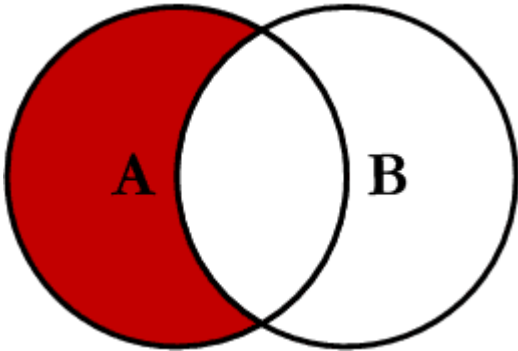
# Outer JOIN



This Join can also be referred to as a FULL OUTER JOIN or a FULL JOIN. This query will return all of the records from both tables, joining records from the left table (table A) that match records from the right table (table B). This Join is written as follows:

```
SELECT <select_list>  
FROM Table_A A  
FULL OUTER JOIN Table_B B  
ON A.Key = B.Key
```

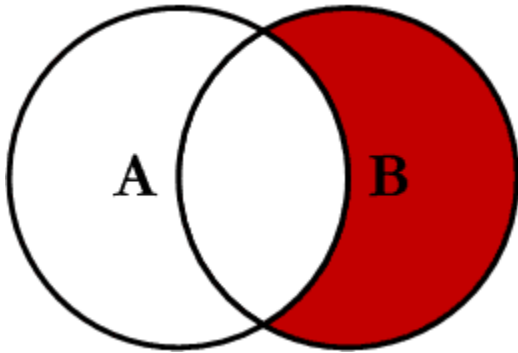
# Left Excluding JOIN



This query will return all of the records in the left table (table A) that do not match any records in the right table (table B). This Join is written as follows:

```
SELECT <select_list>  
FROM Table_A A  
LEFT JOIN Table_B B  
ON A.Key = B.Key  
WHERE B.Key IS NULL
```

# Right Excluding JOIN

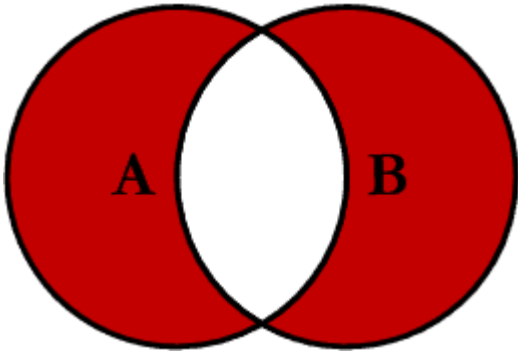


This query will return all of the records in the right table (table B) that do not match any records in the left table (table A). This Join is written as follows:

```
SELECT <select_list>  
FROM Table_A A  
RIGHT JOIN Table_B B  
ON A.Key = B.Key  
WHERE A.Key IS NULL
```



# Outer Excluding JOIN



This query will return all of the records in the left table (table A) and all of the records in the right table (table B) that do not match. I have yet to have a need for using this type of Join, but all of the others, I use quite frequently. This Join is written as follows:

```
SELECT <select_list>  
FROM Table_A A  
FULL OUTER JOIN Table_B B  
ON A.Key = B.Key  
WHERE A.Key IS NULL OR B.Key IS NULL
```

# Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |

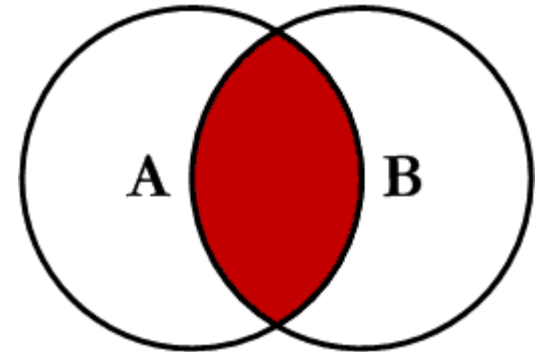
# INNER JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |



```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
INNER JOIN Table_B B  
ON A.PK = B.PK
```

# INNER JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |

| A_PK | A_Value    | B_Value  | B_PK |
|------|------------|----------|------|
| 1    | FOX        | TROT     | 1    |
| 2    | COP        | CAR      | 2    |
| 3    | TAXI       | CAB      | 3    |
| 6    | WASHINGTON | MONUMENT | 6    |
| 7    | DELL       | PC       | 7    |

```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
INNER JOIN Table_B B  
ON A.PK = B.PK
```

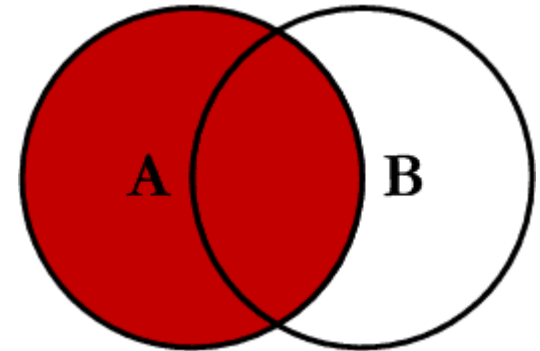
# LEFT JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |



```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
LEFT JOIN Table_B B  
ON A.PK = B.PK
```

# LEFT JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |

| A_PK | A_Value    | B_Value  | B_PK |
|------|------------|----------|------|
| 1    | FOX        | TROT     | 1    |
| 2    | COP        | CAR      | 2    |
| 3    | TAXI       | CAB      | 3    |
| 4    | LINCOLN    | NULL     | NULL |
| 5    | ARIZONA    | NULL     | NULL |
| 6    | WASHINGTON | MONUMENT | 6    |
| 7    | DELL       | PC       | 7    |
| 10   | LUCENT     | NULL     | NULL |

```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
LEFT JOIN Table_B B  
ON A.PK = B.PK
```

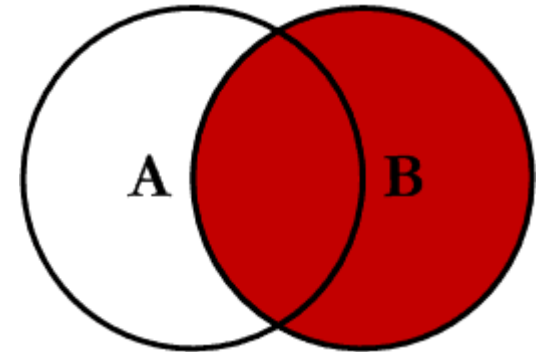
# RIGHTJOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |



```
SELECT  A.PK  AS  A_PK,  A.Value  AS  A_Value,  B.Value  AS
B_Value, B.PK AS B_PK
FROM Table_A A
RIGHT JOIN Table_B B
ON A.PK = B.PK
```

# RIGHTJOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |

| A_PK | A_Value    | B_Value   | B_PK |
|------|------------|-----------|------|
| 1    | FOX        | TROT      | 1    |
| 2    | COP        | CAR       | 2    |
| 3    | TAXI       | CAB       | 3    |
| 6    | WASHINGTON | MONUMENT  | 6    |
| 7    | DELL       | PC        | 7    |
| NULL | NULL       | MICROSOFT | 8    |
| NULL | NULL       | APPLE     | 9    |
| NULL | NULL       | SCOTCH    | 11   |

```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
RIGHT JOIN Table_B B  
ON A.PK = B.PK
```



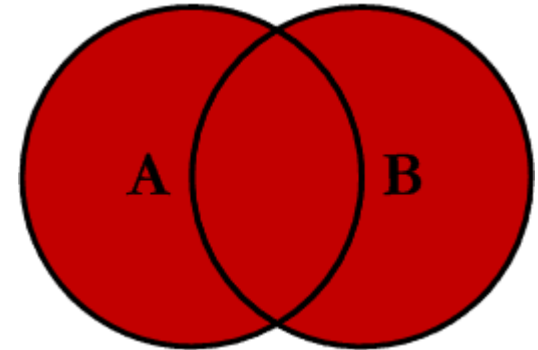
# OUTER JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |



```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
FULL OUTER JOIN Table_B B  
ON A.PK = B.PK
```

# OUTER JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |

| A_PK | A_Value    | B_Value   | B_PK |
|------|------------|-----------|------|
| 1    | FOX        | TROT      | 1    |
| 2    | COP        | CAR       | 2    |
| 3    | TAXI       | CAB       | 3    |
| 4    | LINCOLN    | NULL      | NULL |
| 5    | ARIZONA    | NULL      | NULL |
| 6    | WASHINGTON | MONUMENT  | 6    |
| 7    | DELL       | PC        | 7    |
| 10   | LUCENT     | NULL      | NULL |
| NULL | NULL       | MICROSOFT | 8    |
| NULL | NULL       | APPLE     | 9    |
| NULL | NULL       | SCOTCH    | 11   |

```

SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS
B_Value, B.PK AS B_PK
FROM Table_A A
FULL OUTER JOIN Table_B B
ON A.PK = B.PK
    
```

# OUTER JOIN - Example

MySQL don't have FULL JOINS, but they can be emulated:

```
SELECT  A.PK  AS  A_PK,  A.Value  AS  A_Value,  B.Value  AS  
B_Value, B.PK AS B_PK
```

```
FROM Table_A A
```

```
LEFT JOIN Table_B B
```

```
ON A.PK = B.PK
```

```
UNION
```

```
SELECT  A.PK  AS  A_PK,  A.Value  AS  A_Value,  B.Value  AS  
B_Value, B.PK AS B_PK
```

```
FROM Table_A A
```

```
RIGHT JOIN Table_B B
```

```
ON A.PK = B.PK
```

# OUTER JOIN - Example

UNION vs UNION ALL  
(include duplicates)

| A_PK | A_Value             | B_Value   | B_PK |
|------|---------------------|-----------|------|
| 1    | FOX                 | TROT      | 1    |
| 2    | COP                 | CAR       | 2    |
| 3    | TAXI                | CAB       | 3    |
| 4    | LINCOLN             | NULL      | NULL |
| 5    | ARIZONA             | NULL      | NULL |
| 6    | WASHINGTON MONUMENT |           | 6    |
| 7    | DELL                | PC        | 7    |
| 10   | LUCENT              | NULL      | NULL |
| NULL | NULL                | MICROSOFT | 8    |
| NULL | NULL                | APPLE     | 9    |
| NULL | NULL                | SCOTCH    | 11   |

| A_PK | A_Value             | B_Value   | B_PK |
|------|---------------------|-----------|------|
| 1    | FOX                 | TROT      | 1    |
| 2    | COP                 | CAR       | 2    |
| 3    | TAXI                | CAB       | 3    |
| 4    | LINCOLN             | NULL      | NULL |
| 5    | ARIZONA             | NULL      | NULL |
| 6    | WASHINGTON MONUMENT |           | 6    |
| 7    | DELL                | PC        | 7    |
| 10   | LUCENT              | NULL      | NULL |
| 1    | FOX                 | TROT      | 1    |
| 2    | COP                 | CAR       | 2    |
| 3    | TAXI                | CAB       | 3    |
| 6    | WASHINGTON MONUMENT |           | 6    |
| 7    | DELL                | PC        | 7    |
| NULL | NULL                | MICROSOFT | 8    |
| NULL | NULL                | APPLE     | 9    |
| NULL | NULL                | SCOTCH    | 11   |

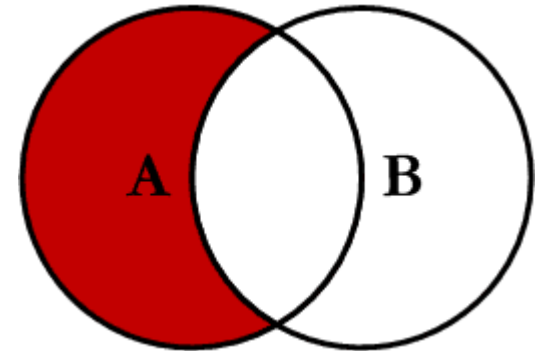
# LEFT EXCLUDING JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |



```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
LEFT JOIN Table_B B  
ON A.PK = B.PK WHERE B.PK IS NULL
```

# LEFT EXCLUDING JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |

| A_PK | A_Value | B_Value | B_PK |
|------|---------|---------|------|
| 4    | LINCOLN | NULL    | NULL |
| 5    | ARIZONA | NULL    | NULL |
| 10   | LUCENT  | NULL    | NULL |

```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
LEFT JOIN Table_B B  
ON A.PK = B.PK WHERE B.PK IS NULL
```

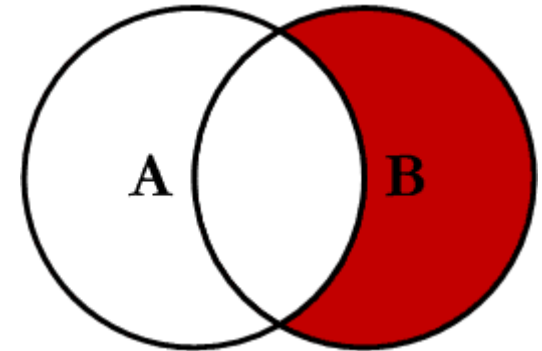
# RIGHT EXCLUDING JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |



```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
RIGHT JOIN Table_B B  
ON A.PK = B.PK WHERE A.PK IS NULL
```

# RIGHT EXCLUDING JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |

| A_PK | A_Value | B_Value   | B_PK |
|------|---------|-----------|------|
| NULL | NULL    | MICROSOFT | 8    |
| NULL | NULL    | APPLE     | 9    |
| NULL | NULL    | SCOTCH    | 11   |

```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
RIGHT JOIN Table_B B  
ON A.PK = B.PK WHERE A.PK IS NULL
```



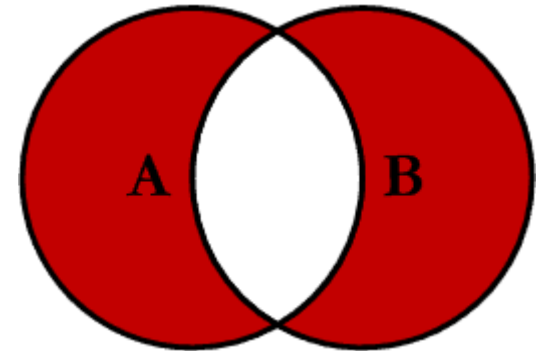
# OUTER EXCLUDING JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |



```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
FULL OUTER JOIN Table_B B  
ON A.PK = B.PK WHERE A.PK IS NULL OR B.PK IS NULL
```

# OUTER EXCLUDING JOIN - Example

TABLE\_A

| PK | VALUE      |
|----|------------|
| 1  | FOX        |
| 2  | COP        |
| 3  | TAXI       |
| 4  | LINCOLN    |
| 5  | ARIZONA    |
| 6  | WASHINGTON |
| 7  | DELL       |
| 10 | LUCENT     |

TABLE\_B

| PK | VALUE     |
|----|-----------|
| 1  | TROT      |
| 2  | CAR       |
| 3  | CAB       |
| 6  | MONUMENT  |
| 7  | PC        |
| 8  | MICROSOFT |
| 9  | APPLE     |
| 11 | SCOTCH    |

| A_PK | A_Value | B_Value   | B_PK |
|------|---------|-----------|------|
| 4    | LINCOLN | NULL      | NULL |
| 5    | ARIZONA | NULL      | NULL |
| 10   | LUCENT  | NULL      | NULL |
| NULL | NULL    | MICROSOFT | 8    |
| NULL | NULL    | APPLE     | 9    |
| NULL | NULL    | SCOTCH    | 11   |

```
SELECT A.PK AS A_PK, A.Value AS A_Value, B.Value AS  
B_Value, B.PK AS B_PK  
FROM Table_A A  
FULL OUTER JOIN Table_B B  
ON A.PK = B.PK WHERE A.PK IS NULL OR B.PK IS NULL
```

# OUTER EXCLUDING JOIN - Example

MySQL don't have FULL JOINS, but they can be emulated:

```
SELECT  A.PK  AS  A_PK,  A.Value  AS  A_Value,  B.Value  AS  
B_Value, B.PK AS B_PK
```

```
FROM Table_A A
```

```
LEFT JOIN Table_B B
```

```
ON A.PK = B.PK WHERE A.PK IS NULL OR B.PK IS NULL
```

```
UNION
```

```
SELECT  A.PK  AS  A_PK,  A.Value  AS  A_Value,  B.Value  AS  
B_Value, B.PK AS B_PK
```

```
FROM Table_A A
```

```
RIGHT JOIN Table_B B
```

```
ON A.PK = B.PK WHERE A.PK IS NULL OR B.PK IS NULL
```